

Positive Sequence Components:

These are the components equal in magnitude and 120° phase displacement. The phase sequence of these components is same as that of original phase sequence of network. The sequence at which the system is working before fault is the original sequence.

Negative phase sequence components:

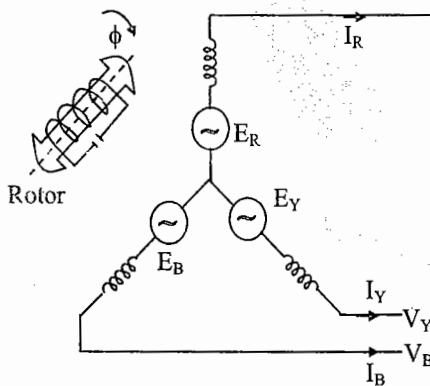
These are the components equal in magnitude 120° phase displacement. The phase sequence of these components opposite to that of original phase sequence of network.

Zero phase sequence components:

These are components are equal in magnitude and without any phase displacement. If there is no phase displacement there is no phase sequence to compare.

Sequence: Direction of rotation of a rotor of a synchronous generator is the reference to decide the sequence.

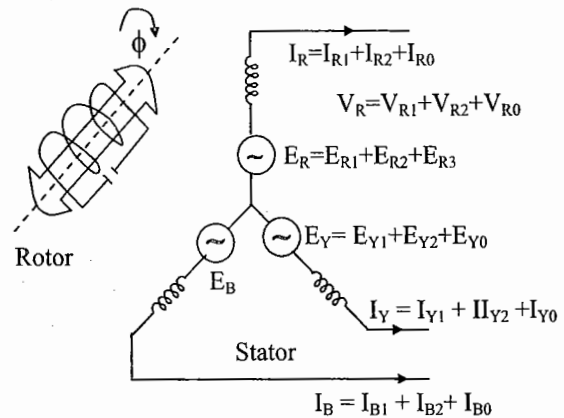
Original Sequence:



It is assumed that the direction of rotation of rotor flux clockwise, so that there will be Clock wise induced emf in stator, which is having a sequence of RYB (or) ABC. The flux produced by stator current is also having direction of rotation in a clock wise manner, which is same as the direction of rotation rotor flux. The corresponding electrical quantities are called Positive sequence components and steady state condition.

The reactance offered is Positive sequence steady state reactance i.e., x_d

$$\begin{aligned} V_R &= V_{R1} & I_R &= I_{R1} & E_R &= E_{R1} \\ V_B &= V_{B1} & I_Y &= I_{Y1} & E_Y &= E_{Y1} \\ V_Y &= V_{Y1} & I_B &= I_{B1} & E_B &= E_{B1} \end{aligned}$$

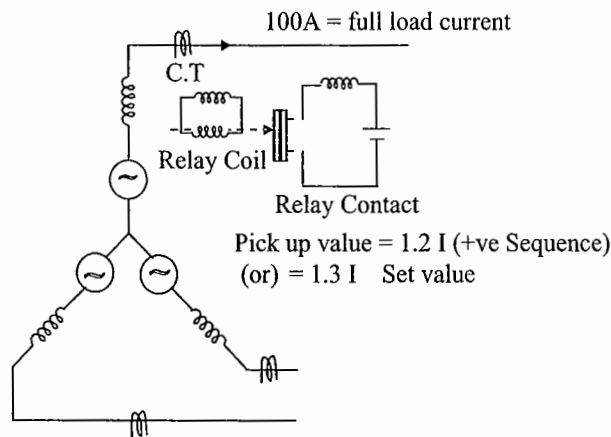


The occurrence of fault will make the system unbalance and it will be represented by three symmetrical components, out of the three components one of the component is +ve sequence component. i.e. the sequence which exist before fault will also exist during the fault but sub transient condition.

There is an unbalance in stator winding but the direction rotation of rotor flux will be always in clock wise manner i.e. same direction as that of before fault. The rotating flux of rotor will be able to produce induced emf in stator in clock wise manner. The flux produced by corresponding current is having direction of rotation same as that of direction of rotation of rotor flux and corresponding electrical quantities are called +ve sequence components sub transient condition. The reactance offered is called x_d'' .

Significance of Positive sequence components:

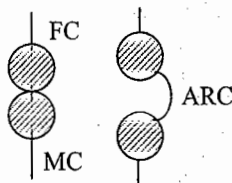
The Positive sequence components of components which are available before fault and also during any type of fault hence these components are bases for relay operation and circuit breaker operation.



The pickup value of relay will be expressed in terms of positive sequence steady state i.e. pickup current = $1.2I$ (or) $1.3I$

During the fault same sequence will exist in sub transient manner so that the relay will operate because the positive sequence sub transient current is more than positive sequence steady state current.

The relays are provided with min pick up value based on the line to ground fault so that it will operate all other faults.



During the normal operation the C.B. contacts are closed. If a fault occurs the moving contact is separated from fixed contact, so that an arc is developed. The circuit is isolated unless arc is break. The breaking of arc result as heat release. The amount of heat release depends upon breaking current.

The heat released by arc should with stand by C.B. contacts. The rating of CB depends on three phase fault of Positive sequence component.

Negative sequence components:

Whenever the direction of stator flux is opposite to that of rotor flux the corresponding electrical

quantities are called negative sequence components R, B, Y. (OR) anticlockwise components

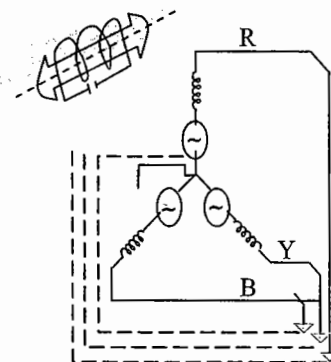
During the fault there will be a terminal voltage at fault point due to unbalance i.e. V_{R_2} but there is no induced voltage in the stator because there is no anti clock wise direction of rotor i.e. E_{R_2} will be zero. Hence the corresponding current from the fault point towards the rotor. The flux produced by the correspond current opposite to that a direction of the rotor flux and corresponding components are called negative sequence components.

Significance of negative sequence components:

The flux produced by negative sequence component is having speed of $-N_s$ however with respect to rotor, it has relative speed of $2N_s$. So that negative sequence flux will able to cuts the rotor field winding which will able to produce an induced emf. Induced emf will once again produce circulating current in rotor field winding with double the frequency. The circulating current will result as the rotor field winding gets over heated. In order to protect the rotor due to over heating negative sequence relay is employed in a stator.

Zero sequence components:

These zero sequence components will exist provided that fault must be ground fault and nearest neutral of the system should be grounded.

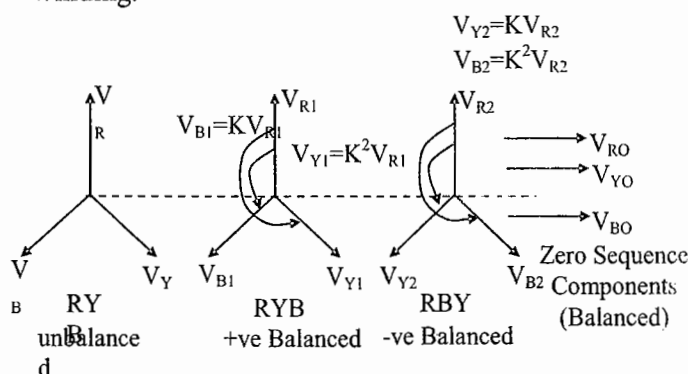


In case of grounded fault the current passes through ground and enter into the system from nearest neutral ground. The ground can provide the magnitude with out any phase. The reactance offered is called x_0 .



Significance of zero sequence components:

The flux produced by the zero sequence component current are equally magnitude without any phase displacement. If there is no phase displacement there is no rotation of flux. So that the effect of zero sequence component flux will not be there on the rotor field winding. However these components are in the form of leakage components in stator winding.



$$\begin{aligned} V_R &= V_{R0} + V_{R1} + V_{R2} = V_{R0} + V_{R1} + V_{R2} \\ V_Y &= V_{Y0} + V_{Y1} + V_{Y2} = V_{R0} + K^2 V_{R1} + K V_{R2} \\ V_B &= V_{B0} + V_{B1} + V_{B2} = V_{R0} + K V_{R1} + K^2 V_{R2} \end{aligned}$$

$$\begin{bmatrix} V_R \\ V_Y \\ V_B \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & K^2 & K \\ 1 & K & K^2 \end{bmatrix} \begin{bmatrix} V_{R0} \\ V_{R1} \\ V_{R2} \end{bmatrix}$$

↓ ↓
un known known
un balanced balanced

The advantage of symmetrical components is to reduce time taking while evaluating unbalanced electrical quantities with help of K notation.

$$\begin{bmatrix} I_R \\ I_Y \\ I_B \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & K^2 & K \\ 1 & K & K^2 \end{bmatrix} \begin{bmatrix} I_{R0} \\ I_{R1} \\ I_{R2} \end{bmatrix}$$

If a fault occurs, the system is unbalanced and use unbalanced quantities to calculate balanced symmetrical components.

$$\begin{bmatrix} I_{R0} \\ I_{R1} \\ I_{R2} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & K^2 & K \\ 1 & K & K^2 \end{bmatrix}^{-1} \begin{bmatrix} I_R \\ I_Y \\ I_B \end{bmatrix}$$

$$\begin{bmatrix} I_{R0} \\ I_{R1} \\ I_{R2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & K & K^2 \\ 1 & K^2 & K \end{bmatrix} \begin{bmatrix} I_R \\ I_Y \\ I_B \end{bmatrix}$$

The symmetrical components are associated with their networks which are called as symmetrical networks (or) sequence networks.

Symmetrical Networks (or) sequence networks: (Per phase Sub-Transient Impedance (Reactance) networks)

In case of positive and negative sequence network, neutral is the reference. In case of zero components ground is reference.

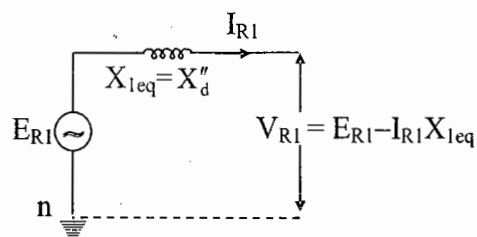
$$I_n = I_R + I_Y + I_B \neq 0 \text{ (un balanced)}$$

$$\begin{aligned} I_n &= I_{R0} + I_{R1} + I_{R2} + I_{Y0} + I_{Y1} + I_{Y2} + I_{B0} + I_{B1} + I_{B2} \\ &= I_{R0} + I_{R1} + I_{R2} + I_{R0} + K^2 I_{R1} + I_{R2} + I_{R0} + K I_{R1} + K^2 I_{R2} \\ &= 3I_{R0} + I_{R1}(1 + K^2 + K) + I_{R2}(1 + K + K^2) \\ &= 3I_{R0} \end{aligned}$$

$$I_n = 3I_{R0}$$

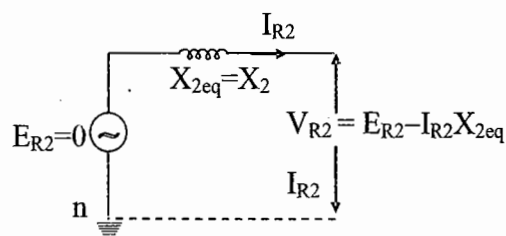
In positive and negative sequence the effect of neutral grounding is not required to include. In case of zero sequence the effect neutral grounding to be included.

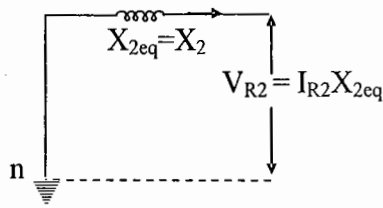
Positive sequence Network:



$V_{R1} = E_{R1} - I_{R1} X_{1eq}$ is called positive sequence Network Terminal voltage equation.

Negative sequence Network:





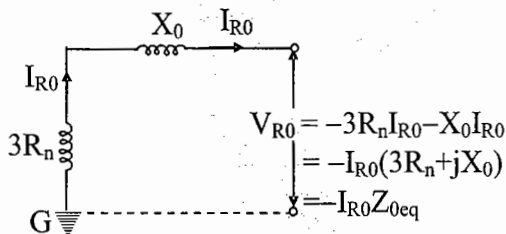
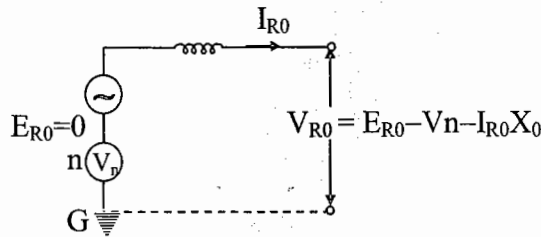
$V_{R2} = -I_{R2}X_{2eq1}$ is called as Negative sequence network terminal voltage equation.

Zero sequence Network:

Consider the effect of neutral grounding, because the reference is ground. The effect of neutral grounding is in terms of neutral voltage

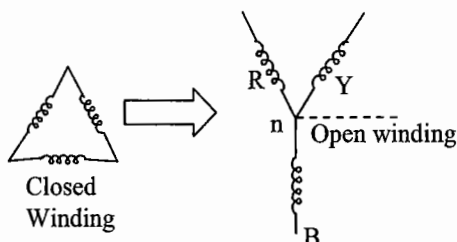
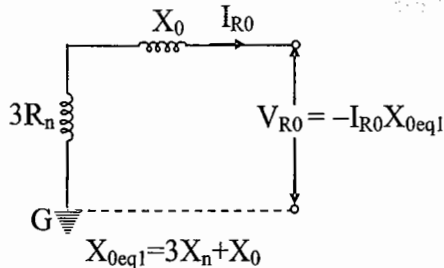
$$V_n = I_n R_n \quad (\text{Resistance grounding})$$

$$V_n = 3I_{R0}R_n \text{ with negative sign}$$



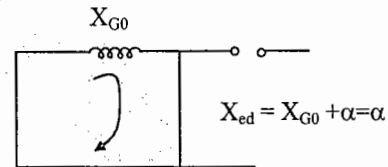
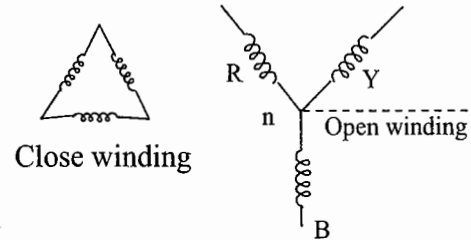
$$V_{R0} = -I_{R0}Z_{0eq1}$$

Zero sequence network terminal voltage equation
Reactance Neutral grounding:



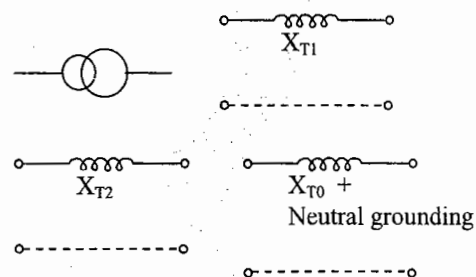
In a Δ connected winding the positive & negative sequence component exists because the reference is neutral. The zero sequence components does not exist the network not connected to ground.

Y-winding is called open winding and phase currents are same as line current.

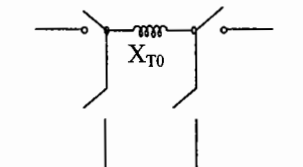


In case of Δ connected zero sequence currents are circulated with in the phases but currents are not entered to line because there is no grounding of system.

Zero sequence networks of T/Fs:



In order to include neutral grounding effect of T/F. The T/F can be represented mechanical equivalent as series parallel switches.



Series winding $\rightarrow$ Y winding

Parallel wind $\rightarrow$ Δ winding

Series Switch - Y (open winding)

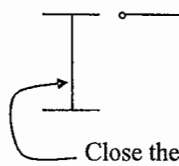
Parallel Switch - Δ (closed winding)



— No closed path — Open — $3X_n = \alpha$

— Closed path — Close — $3X_n = 0$

— Closed path — Close — $3X_n$



Close the parallel switch

S.No	Connection	Impedance/Reactance
1		
2		
3		
4		
5		

6		
7		
8		
9		
10		
11		

name	+ve Sequence Reactance	-ve sequence Reactance	Zero sequence Reactance
Tr. line	X_1	$X_2 = X_1$	$X_0 = 3X_1$
Transformer	X_1	$X_2 = X_1$	$X_0 = X_1$
Cylindrical Rotor Syn Machine	X_d''	$X_2 = X_d''$	$X_0 \ll X_d''$
Salient pole Rotor syn machine	X_d''	$X_2 < X_d''$	$X_0 \ll X_d''$



The negative sequence reactance is same as that of positive sequence reactance except in a salient pole synchronous machine due to non uniform distribution of flux in the air gap due to projected poles. The negative sequence reactance is the average reactance of direct axis reactance and quadrature axis reactance.

$$X_2 = \frac{X_d'' + X_q''}{2}$$

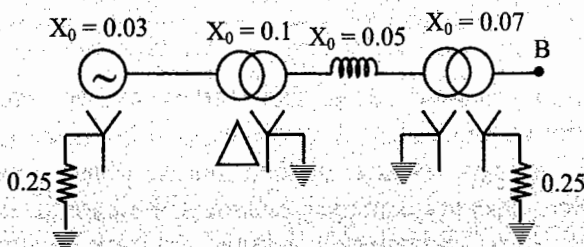
$$X_2 < X_d''$$

The zero sequence components are present when the fault is a ground fault. In any ground fault, the current passes through ground, so that the effect of ground impedance (OR) reactance is to be considered in a longer length of equipments like transmission line and the effect of ground is ignored in a shorter length of equipments. If the ground effect is considered in transmission line, the zero sequence reactance is a variable value due to different soils and change in moisture in the ground. The value is varied as $2.5X_1$ to $3.5X_1$ and an average of $3X_1$.

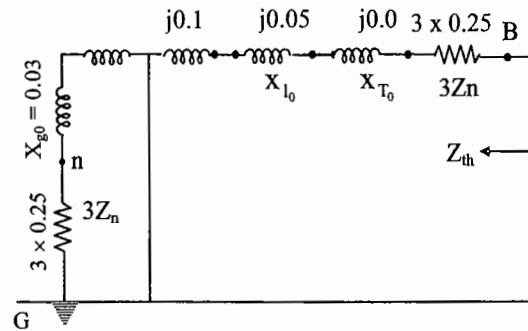
In case of synchronous machine, the zero sequence reactance is treated as leakage reactance which is very less when compared to positive sequence reactance because armature reaction effect is zero so that $X_a = 0$

Example 3.3

A generator is connected to a transformer which feeds another transformer through a short feeder. The zero sequence impedance values are expressed in pu on a common base and are indicated in figure. The Thevenin's equivalent zero sequence impedance at point B is



Sol: Per unit zero sequence reactance diagram of the given single line diagram is shown below.

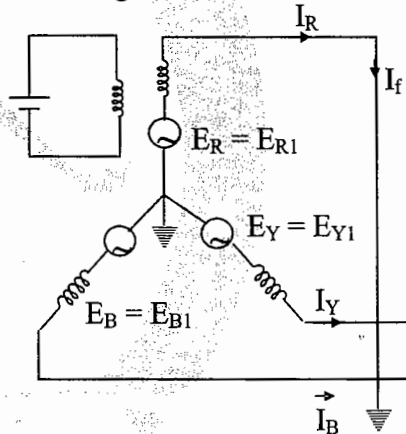


Thevenin's equivalent impedance, Z_{th} at 'B' is

$$\begin{aligned} Z_{th} &= j0.1 + j0.05 + j0.07 + 0.75 \\ &= 0.75 + j0.22 \end{aligned}$$

Fault analysis:

Line to ground fault: Alternator is working at no load, rated voltage solid neutral and solid fault.



$$\left. \begin{aligned} I_f &= I_R \\ I_Y &= I_B = 0 \\ V_R &= 0 \end{aligned} \right\} \text{Un balanced}$$

Use the symmetrical components for fault analysis.

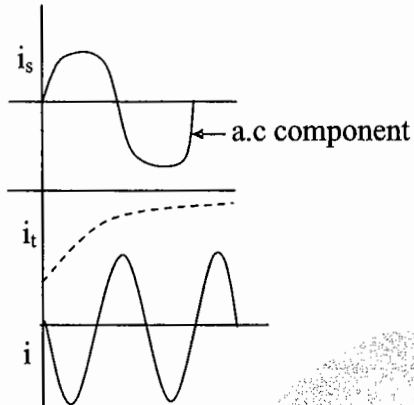
$$I_{R0} = \frac{1}{3} [I_R + I_Y + I_B] = \frac{I_R}{3}$$

$$I_{R1} = \frac{1}{3} [I_R + KI_Y + K^2I_B] = \frac{I_R}{3}$$

$$I_{R2} = \frac{1}{3} [I_R + K^2I_Y + KI_B] = \frac{I_R}{3}$$

$$I_{R0} = I_{R1} = I_{R2} = \frac{I_R}{3}$$

The fault is unbalanced with ground so that the fault is having three components in any grounded fault, the fault current should be in terms of zero sequence currents.



$$I_f = I_R = 3I_{R0} \text{ P.U.}$$

$$I_f = I_R = 3I_{R1} \text{ P.U.}$$

$$I_{R0} = I_{R1}$$

The fault current which is calculated symmetrical sinusoidal current, that will be expressed in terms of r.m.s. value.

$$I_{f(\text{PU})} = \frac{I_f(a)}{I_b}$$

$$I_f(a) = I_f(\text{P.U.}) I_b \text{ A or KA}$$

$$S = \sqrt{3} V_L I_L$$

$$I_L = \frac{S}{\sqrt{3} V_L}$$

$$I_P = I_L = I_b \text{ (Y system)}$$

Calculation of I_{R1} :

$$V_R = 0$$

$$V_{R1} + V_{R2} + V_{R0} = 0$$

$$E_{R1} - I_{R1} X_{1eq} - I_{R2} X_{2eq} - I_{R0} X_{0eq} = 0$$

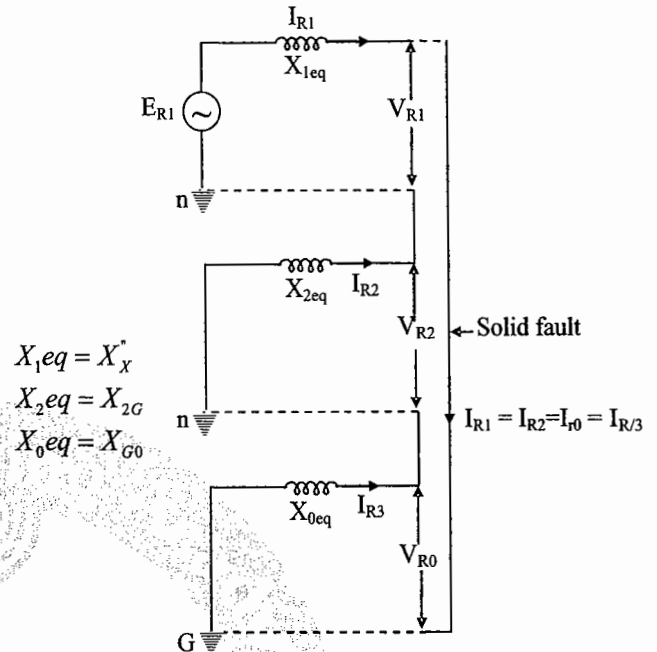
$$E_{R1} = I_{R1} X_{1eq} - I_{R2} X_{2eq} + I_{R0} X_{0eq}$$

$$(I_{R1} = I_{R2} = I_{R3})$$

$$I_{R1} = \frac{E_{R1}}{X_{1eq} + X_{2eq} + X_{0eq}} \text{ positive sequence current}$$

$$I_f = 3I_{R1} = \frac{3E_{R1}}{X_{1eq} + X_{2eq} + X_{0eq}}$$

The fault is having 3 components so that there will be three networks and the networks are connected in series because the sequence currents are same.



The rating of C.B. will be expressed in terms of braking current. This is called as fault current (or) short circuit current. The rating of circuit breaker will also expressed breaking capacity or short circuit capacity = $3V_b I_f$

Potential of neutral:

$$V_n = I_n X_n = 3I_{R0} X_n = I_f X_n \text{ P.U.}$$

In any grounded fault the current through neutral (or) current through ground is same as fault current only.

$$V_n(a) = V_n \text{ P.U.} \times V_b \text{ volts (or) kV.}$$

$$V_n = 0 \text{ for solid neutral.}$$

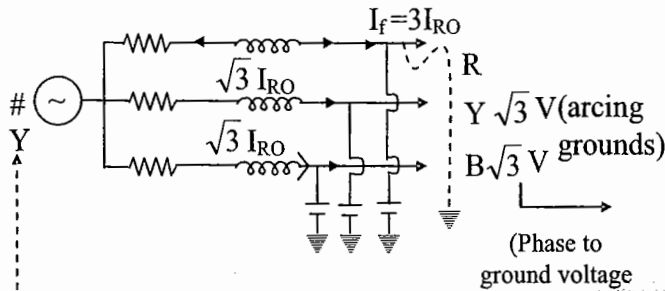
$$\text{Isolated neutral } X_{G0} + \alpha = \alpha$$

$$I_f = 3I_{R1} = \frac{3XE_{R1}}{X_{1eq} + X_{2eq} + \alpha} = 0 \text{ P.U.}$$

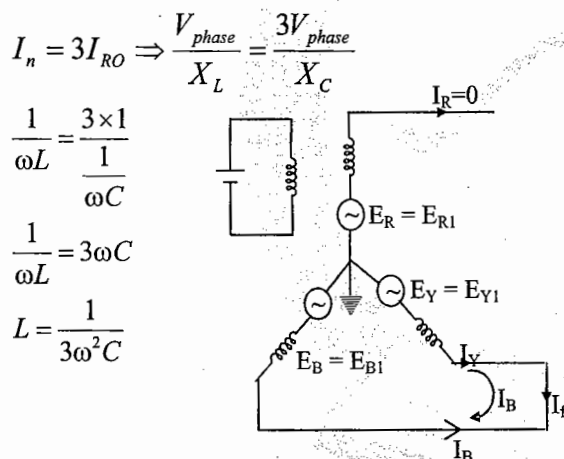
In case of Isolated neutral fault current will be zero and the voltages of healthy phases will be same as rated voltage. Hence neither faulty phase nor healthy phase not damaged. However if alternator is connected to transmission line and the fault occurs at end of the line, the neutral of alternator is Isolated. The fault current is zero but the voltages of healthy phase become $\sqrt{3}V$ which is called as



arcing ground. The insulation of phase is selected for $\sqrt{2}V$. So that dielectric strength of insulation failed in healthy phases. The arcing ground due to capacitance of line.



In order to eliminate arcing grounds the neutral of the system to be connected to ground through a coil which is known as Peterson Coil. The type of grounding is called resonance grounding. The value of inductance of coil $L = \frac{1}{3\omega^2 C}$



Example 3.4:

A 10 MVA, 13.8 kV alternator has positive, negative and zero sequence reactance of 30%, 40%, and 5% respectively. What reactance must be put in the generator neutral so that the fault current for a line to ground fault will not exceed the rated current.

Sol: Fault current = Rated current

$$I_{d.p.u} = 1.0 \text{ p.u.}$$

$$1.0 = \frac{3E_{R_1}}{X_1 + X_2 + X_0 + 3X_n}$$

$$1.0 (X_1 + X_2 + X_0 + 3X_n) = 3$$

$$0.3 + 0.4 + 0.05 + 3X_n = 3$$

$$X_n = 0.75 \text{ p.u.}$$

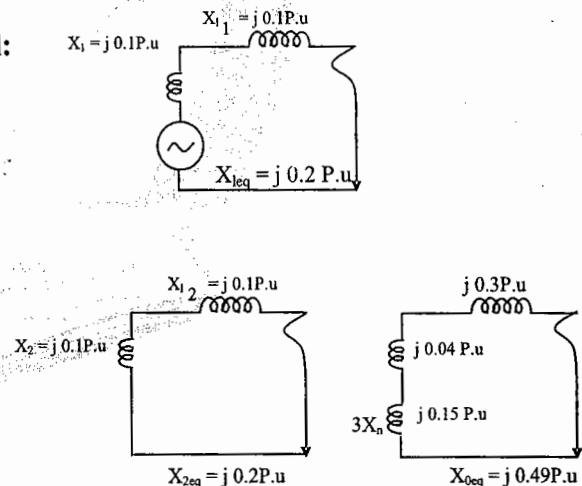
$$X_{n(\Omega)} = 0.75 \left(\frac{KV_b^2}{MVA_b} \right)$$

$$= 0.75 \left[\frac{13.8^2}{10 \text{ MVA}} \right] = 14.28 \Omega$$

Example 3.5

A 20-MVA, 6.6-kV, 3-phase alternator is connected to a 3-phase transmission line. The per unit positive-sequence, negative-sequence and zero-sequence impedance of the alternator are $j0.1$, $j0.1$ and $j0.04$ respectively. The neutral of the alternator is connected to ground through an inductive reactor of $j0.05$ p.u. The per unit positive-, negative- and zero-sequence impedances of the transmission line are $j0.1$, $j0.1$ and $j0.3$, respectively. All per unit values are based on the machine ratings. A solid ground fault occurs at one phase of the far end of the transmission line. The voltage of the alternator neutral with respect to ground during the fault is

Sol:



The fault current for line to ground fault

$$I_f = \frac{3E_{R_1}}{X_{1eq} + X_{2eq} + X_{0eq}}$$

Where, $E_{R_1} = V_{th} = 1.0 \text{ p.u.}$

$$I_f = \frac{3 \times 1.0}{j0.2 + j0.2 + j0.49} = 3.37 \angle -90^\circ$$



$$\begin{aligned}\text{The voltage of neutral reactance} &= I_n X_n \\ &= I_f X_n \\ &= 3.37 \times 0.05 = 0.168 \text{ p.u.}\end{aligned}$$

Since the voltage of alternator is taken as base, neutral reactance

$$\text{Voltage, } V_n = 0.168 \times \frac{6600}{\sqrt{3}} = 642.2 \text{ V}$$

L-L fault:

During Fault:

Unbalanced condition

$$I_f = I_Y = -I_B$$

$$I_Y + I_B = 0$$

$$I_R = 0$$

$$V_Y = V_B$$

Use the symmetrical components

$$I_{R0} = \frac{1}{3} [I_R + I_Y + I_B] = 0$$

$$I_{R1} = \frac{1}{3} [I_R + KI_Y + K^2 I_B]$$

$$= \frac{1}{3} [0 + KI_Y - K^2 I_B]$$

$$= \frac{I_Y}{3} [K - K^2]$$

$$I_{R2} = \frac{1}{3} [I_R + K^2 I_Y + KI_B]$$

$$= \frac{I_Y}{3} [K^2 I_Y - KI_Y] = \frac{I_Y}{3} [K^2 - K]$$

$$= -\frac{I_Y}{3} [K - K^2]$$

$$I_{R2} = -I_{R1} \text{ ----- (i)}$$

The fault is unbalanced fault without ground, there is no zero sequence components the fault is having positive, negative sequence components.

$$V_Y = V_B$$

$$V_{Y0} + V_{Y1} + V_{Y2} = V_{B0} + V_{B1} + V_{B2}$$

$$K^2 V_{R1} + KV_{R2} = KV_{R1} + K^2 V_{R2}$$

$$(K^2 - K)V_{R1} = (K^2 - K)V_{R2}$$

$$V_{R1} = V_{R2} \text{ ----- (ii)}$$

The fault is having two networks and they connected in anti parallel.

Positive sequence current is

$$V_{R1} = V_{R2}$$

$$E_{R1} - I_{R1} X_{1eq} = -I_{R2} X_{2eq}$$

$$E_{R1} = I_{R1}(X_{1eq} + X_{2eq})$$

Calculation of I_f :

$$I_f = I_Y = I_{Y0} + I_{Y1} + I_{Y2}$$

$$I_f = 0 + K^2 I_{R1} + KI_{R2}$$

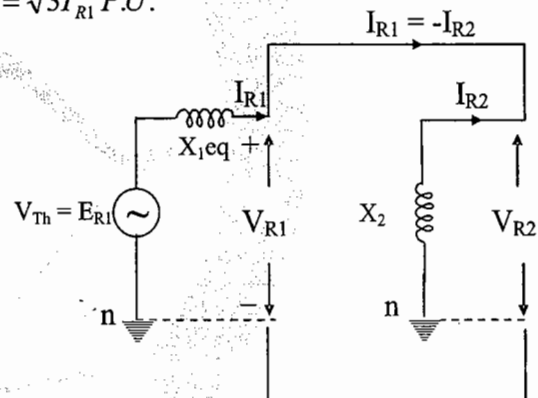
$$I_f = 0 + K^2 I_{R1} - KI_{R1}$$

$$I_f = (K^2 - K) I_{R1}$$

$$I_f = [(-0.5 - j0.867) - (-0.5 + j0.867)]$$

$$I_f = -j1.732 I_{R1} = j1.732 I_{R2}$$

$$I_f = \sqrt{3} I_{R1} \text{ P.U.}$$



$$I_{R1} = \frac{E_{R1}}{X_{1eq} + X_{2eq}} \text{ P.U.}$$

$$I_f = \sqrt{3} I_{R1} \text{ P.U.}$$

$$I_f = \sqrt{3} \times \frac{E_R}{X_{1eq} + X_{2eq}} \text{ P.U.}$$

$$I_f(a) = I_f \text{ P.U.} \times I_b \text{ A (or) K.A.}$$

Short circuit capacity

$$= 3 V_b I_f$$

Potential of neutral

$$V_n = I_n X_n \quad V_n = 3 I_{R0} X_n = 0$$

In case of reactance neutral (or) isolated neutral the fault current remains same. +ve, -ve sequence network does not depend upon neutral grounding for L-L faults.



L-L-G fault:

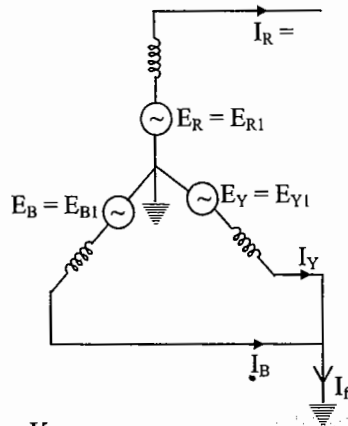
During fault:

Unbalanced

$$I_f = I_Y + I_B$$

$$I_R = 0$$

$$V_Y = V_B = 0$$



$$V_{R0} = \frac{1}{3}[V_R + V_Y + V_B] = \frac{V_R}{3}$$

$$V_{R1} = \frac{1}{3}[V_R + KV_Y + K^2V_B] = \frac{V_R}{3}$$

$$V_{R2} = \frac{1}{3}[V_R + K^2V_Y + KV_B] = \frac{V_R}{3}$$

$$V_{R0} = V_{R1} = V_{R2} = \frac{V_R}{3} \quad \text{----- (i)}$$

The fault is unbalanced with ground the fault is having three components

$$I_f = I_Y + I_B$$

$$I_f = I_{Y0} + I_{Y1} + I_{Y2} + I_{B0} + I_{B1} + I_{B2}$$

$$I_f = I_{R0} + K^2I_{R1} + KI_{R2} + I_{R0} + KI_{R1} + K^2I_{R2}$$

$$I_f = 2I_{R0} + I_{R1}(K^2 + K) + I_{R2}(K + K^2)$$

$$I_f = 2I_{R0} - I_{R1} - I_{R2} \quad \text{----- (ii)}$$

$$I_R = I_{R0} + I_{R1} + I_{R2} = 0$$

$$I_{R0} = I_{R1} - I_{R2}$$

$$I_f = 2I_{R0} + I_{R0} = 3I_{R0} \text{ P.U.}$$

Calculation of I_{R1} :

$$I_{R1} + I_{R2} + I_{R0} = 0$$

Replace I_{R2} & I_{R0} in terms of I_{R1} .

$$V_{R1} = V_{R2}$$

$$E_{R1} - I_{R1}X_{1eq} = -I_{R2}X_{2eq}$$

$$I_{R2} = -\left[\frac{E_{R1} - I_{R1}X_{1eq}}{X_{2eq}}\right] \quad \text{----- (iii)}$$

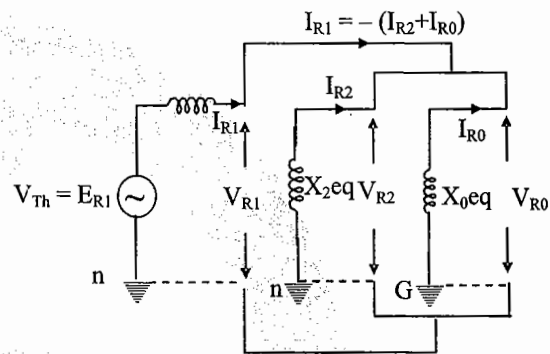
$$V_{R1} = V_{R0}$$

$$E_{R1} - I_{R1}X_{1eq} = -I_{R0}X_{0eq}$$

$$I_{R0} = -\left[\frac{E_{R1} - I_{R1}X_{1eq}}{X_{0eq}}\right] \quad \text{----- (iv)}$$

$$I_{R1} = \frac{E_{R1}}{X_{1eq} + \frac{X_{0eq} + X_{2eq}}{X_{2eq} + X_{0eq}}}$$

The fault is having three networks are connected in the parallel combination negative and zero sequence network are connected in series with positive sequence network.



$$I_{R0} = -I_{R1} \frac{X_{2eq}}{X_{2eq} + X_{0eq}} \text{ P.U. Current Divider Rule.}$$

$$I_f = 3I_{R0} \text{ P.U.}$$

$$I_f(a) = I_f(\text{P.U.}) I_b$$

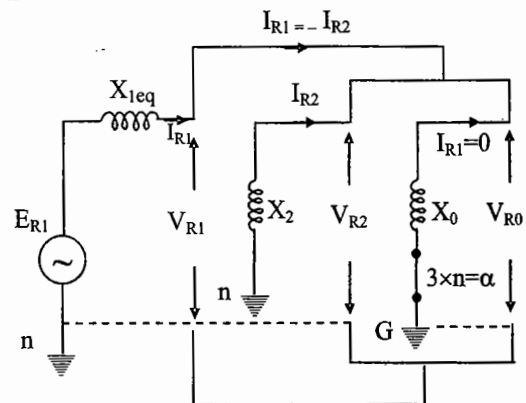
$$\text{Short circuit capacity} = 3V_b I_f$$

$$\text{Potential of neutral } V_n = I_n X_n$$

$$V_n = 3I_{R0}X_n = I_f X_n$$

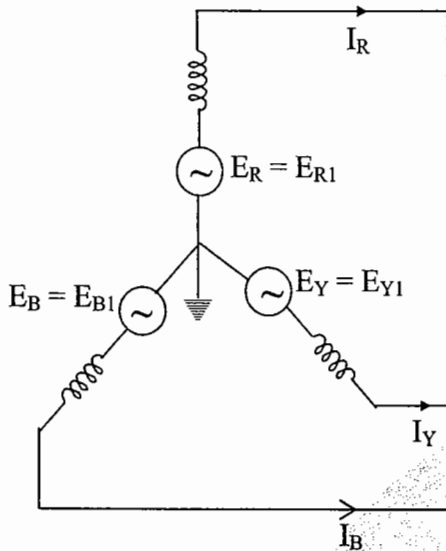
In case of solid neutral $V_n = 0$.

In case of isolated neutral current into the ground is zero there will be current in the phase so that it will be treated as L-L fault.





Line to line to line: (or) balanced



During fault:

$$I_f = I_R = I_Y = I_B$$

$$I_R + I_Y + I_B = 0$$

$$V_R = V_Y = V_B$$

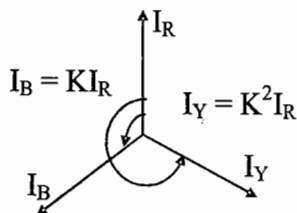
Fault is having only positive set component it can be proved by symmetrical components.

$$I_{R0} = \frac{1}{3} [I_R + I_Y + I_B] = 0$$

$$I_{R1} = \frac{1}{3} [I_R + KI_Y + K^2 I_B] = I_R$$

$$I_{R2} = \frac{1}{3} [I_R + K^2 I_Y + KI_B] = 0$$

$$I_f = I_R = I_{R1} \text{ P.U.} \rightarrow \text{because balance fault.}$$



$$I_{R0} = 0, V_{R0} = 0$$

$$I_{R2} = 0, V_{R2} = 0$$

$$V_{R1} = \frac{1}{3} [V_R + KV_Y + K^2 V_B] = \frac{V_R}{3} (1 + K + K^2)$$

$$V_{R1} = 0$$

In 3 - ϕ short circuit fault the +ve sequence network terminal voltage is zero.

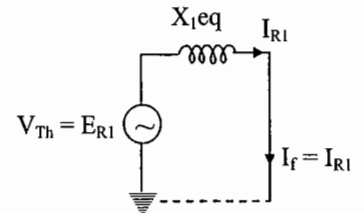
$$V_{R1} = 0$$

$$E_{R1} \times I_{R1} X_{1eq} = 0$$

$$I_{R1} = \frac{E_{R1}}{X_{1eq}} \text{ P.U.}$$

$$I_f = I_R = \frac{E_{R1}}{X_{1eq}} \text{ P.U.}$$

$$I_f(a) = I_f(\text{P.U.}) I_b$$



Short Circuit capacity:

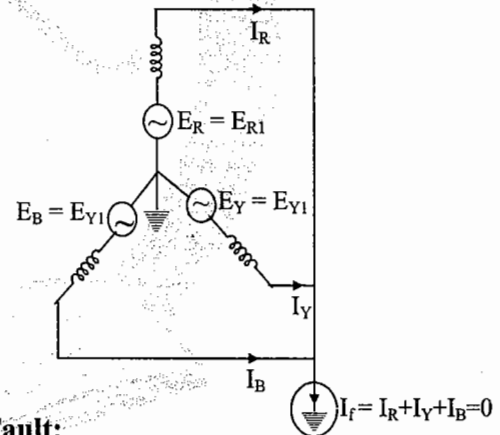
$$3V_b I_f$$

$$V_n = I_n X_n = 3I_{R0} X_n = 0$$

The fault does not depends up on neutral grounding.

Line to line to line to ground:

$$L-L-L-G = L-L-L$$



3- ϕ Fault:

$$I_f = \frac{E_{R1}}{X_{1eq}}$$

$$\text{Short circuit capacity} = 3V_b I_f = 3V_b I_f \text{ P.U. } I_b$$

$$= 3V_b I_b I_f \text{ P.U.}$$

$$= 3V_b I_b \frac{E_{R1}}{X_{1eq}}$$

$$= 3V_b I_b \frac{1.0}{X_{1eq}}$$

$$\text{Short circuit capacity} = \frac{\text{Base MVA}}{X_{1eq}}$$



For synchronous generator

$$X_d'' < X_d' < X_d$$

We know that sub-transient reactance is very less so that sub-transient current is very high and short circuit capacity of fault will be high which will result as the rating of C.B. will be high. Hence, the cost of the breaker will be high. In order to reduce cost of the breaker, the short circuit capacity is reduced by increasing the reactance the network, for which it is proposal to have a series reactor in the circuit.

Example 3.6:

A 20 MVA, 33 kV, 3 – phase alternator is subjected to different types of short circuits and the following are the values of fault current.

3 – phase short circuit – 319A

Single line to ground – 659A

Line to line – 435A

Determine +ve, -ve and zero sequence reactances in p.u.

Sol: $S = \sqrt{3} V_L I_L$

$$I_b = I_L = \frac{S}{\sqrt{3} V_L} = \frac{20 \times 10^6}{\sqrt{3} \times 33 \times 10^3}$$

$$I_b = 349.90 \approx 350 \text{ A}$$

LLL Fault:

$$I_{f(a)} = I_{fpu} \times I_b$$

$$319 = \frac{E_{R_1}}{X_{1pu}} \times I_b$$

$$X_{1pu} = \frac{1}{319} \times 350 = 1.09$$

$$X_{1\Omega} = X_{1pu} \left[\frac{(KV_b)^2}{MVA_b} \right]$$

$$= 1.09 \times \left(\frac{33^2}{20} \right) = 59.35 \Omega$$

$$X_{1\Omega} = 59.35 \Omega$$

LL Fault :

$$I_{f(a)} = I_{fpu} \times I_b$$

$$435 = \frac{\sqrt{3} \times E_{R_1}}{X_{1pu} + X_{2pu}} \times 350$$

$$435 = \frac{\sqrt{3} \times 1}{X_1 + X_2} \times 350$$

$$X_1 + X_2 = 1.393 \text{ pu}$$

$$X_2 = 1.393 - 1.09 = 0.303 \text{ pu}$$

$$X_{2\Omega} = 0.303 \left(\frac{33^2}{20} \right) = 16.49 \Omega$$

$$X_{2\Omega} = 16.49 \Omega$$

LG Fault :

$$I_{f(a)} = I_{fpu} \times I_b$$

$$659 = \frac{3 \times E_{R_1}}{X_1 + X_2 + X_0} \times 350$$

$$X_1 + X_2 + X_0 = 1.59 \text{ pu}$$

$$X_{0pu} = 1.59 \text{ p.u} - (X_1 + X_2)$$

$$= 1.59 \text{ p.u} - (1.09 + 0.303)$$

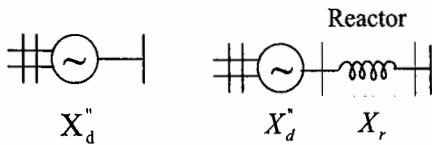
$$= 0.197 \text{ p.u}$$

$$X_{0\Omega} = X_{0pu} \left[\frac{KV_b^2}{MVA_b} \right]$$

$$= 0.197 \left[\frac{33^2}{20} \right] = 10.726 \Omega$$

$$\therefore X_{0\Omega} = 10.726 \Omega$$

1. Generator reactor: Generator is connected to bus through a reactor.

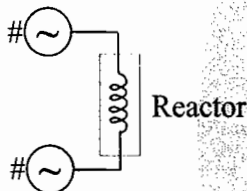


$$I_f = \frac{E_{R1}}{X_x} \quad I_f = \frac{E_{R1}}{X_x'' + X}$$

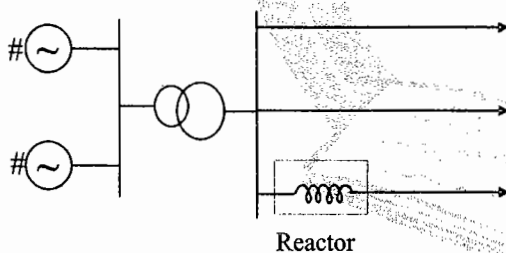
$$SC \text{ MVA} = \frac{\text{Base MVA}}{X_d''}$$

$$SC \text{ MVA} = \frac{\text{Base MVA}}{X_d'' + X_r}$$

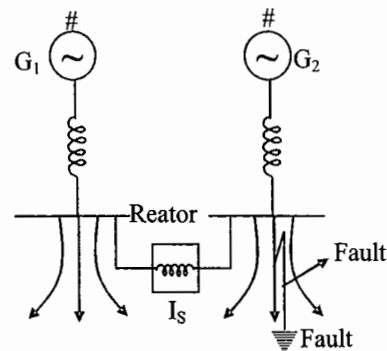
2. Bus bar Reactor:



3. Feeder Reactor:



Inter connected system:



In order to improve the reliability of supply, it is proposed to connect the above two stations as inter connecting stations by providing on inter connector.

Inter connecting network is having one disadvantage i.e. the short circuit current level will be high. If fault occurs any one the area because other area also supply the current in order reduce fault, the since relay is employed.